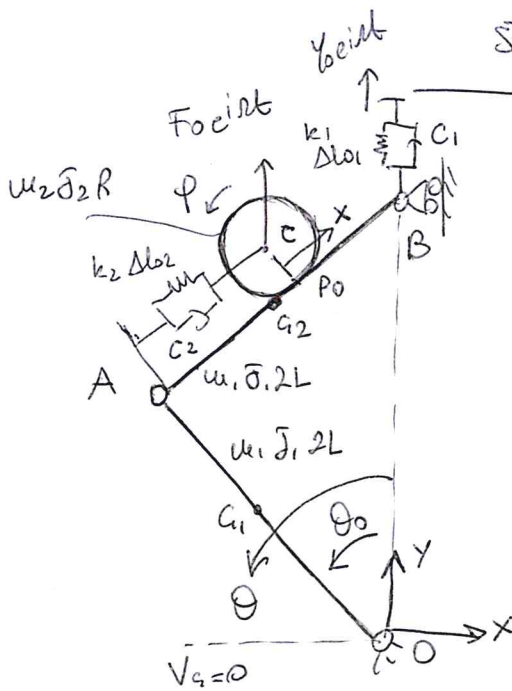


Standard problem



$$\underline{x} = \begin{Bmatrix} \theta \\ x \end{Bmatrix}$$

$$\Delta p_0 = L_0 \quad \underline{x}_0 = \begin{Bmatrix} \theta_0 \\ 0 \end{Bmatrix}$$

$$V_{G1} = L \dot{\theta}$$

$$x_{G2} = -L \sin \theta$$

$$y_{G2} = 3L \cos \theta$$

$$y_B = 4L \cos \theta \rightarrow \Delta L_1 = -(y_B - y_{B0}) = 4L \cos \theta_0 - 4L \cos \theta$$

$$x_C = -2L \sin \theta + (L_0 + x) \sin \theta - R \cos \theta$$

$$y_C = 2L \cos \theta + (L_0 + x) \cos \theta + R \sin \theta$$

$$\Delta L_2 = x; \quad \varphi = -\theta - x/R$$

$$h_{G1} = L \cos \theta; \quad h_{G2} = 3L \cos \theta; \quad h_{G3} = y_C$$

$$\frac{\partial x_{G2}}{\partial \theta} = -L \cos \theta; \quad \frac{\partial y_{G2}}{\partial \theta} = -3L \sin \theta$$

$$\frac{\partial x_C}{\partial \theta} = -2L \cos \theta + (L_0 + x) \cos \theta + R \sin \theta; \quad \frac{\partial x_C}{\partial x} = \sin \theta$$

$$\frac{\partial y_C}{\partial \theta} = -2L \sin \theta - (L_0 + x) \sin \theta + R \cos \theta; \quad \frac{\partial y_C}{\partial x} = \cos \theta$$

$$\frac{\partial \Delta L_1}{\partial \theta} = 4L \sin \theta$$

$$T = \frac{1}{2} m_1 v_{G1}^2 + \frac{1}{2} m_2 v_{G2}^2 + \frac{1}{2} m_1 \dot{x}_{G2}^2 + \frac{1}{2} m_1 \dot{y}_{G2}^2 + \frac{1}{2} m_2 \dot{x}_C^2 + \frac{1}{2} m_2 \dot{y}_C^2 + \frac{1}{2} I_2 \dot{\varphi}^2$$

$$D = \frac{1}{2} \sum_{i=1}^2 C_i \Delta L_i^2; \quad V_{el} = \frac{1}{2} \sum_{i=1}^2 k_i \Delta L_i^2; \quad V_g = m_1 g h_{G1} + m_2 g h_{G2} + m_2 g h_{G3}$$

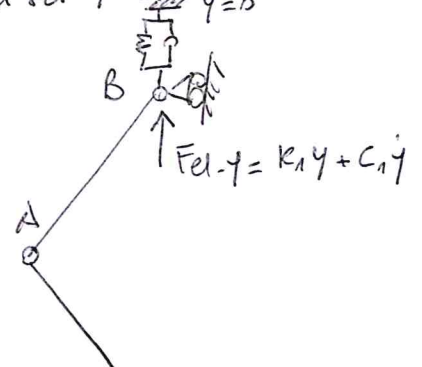
$$\delta W = F_{el,1} \delta y_C + F_{el,2} \delta x_B$$

→ solution applying the equivalent force $F_{el,2}$ at point B and set $y = 0$ at point B

$$[M_{eff}] = \text{diag}(m_1, 2I_2, m_1, m_1, m_2, m_2, I_2)$$

$$[C_{eff}] = \text{diag}(C_1, C_2) \quad K_{eff} = \text{diag}(k_1, k_2)$$

$$F = \begin{Bmatrix} F_{el,1} \\ F_{el,2} \end{Bmatrix}^T$$



$$[\Delta m] = \begin{bmatrix} L & \phi \\ 1 & \phi \\ -L \cos \theta_0 & \phi \\ -3L \sin \theta_0 & \phi \\ \lambda_{u1} & \sin \theta_0 \\ \lambda_{u2} & \cos \theta_0 \\ -1 & -1/R \end{bmatrix}$$

$$\lambda_{u1} = (-2L + L_0) \cos \theta_0 + R \sin \theta_0$$

$$\lambda_{u2} = -(2L + L_0) \sin \theta_0 + R \cos \theta_0$$

$$[\Delta c_k] = [\Delta c] = \begin{bmatrix} 4L \sin \theta_0 & 0 \\ 0 & 1 \end{bmatrix}$$

$$[\Delta e] = \begin{bmatrix} -(2L + L_0) \sin \theta_0 + R \cos \theta_0 & \cos \theta_0 \\ -4L \sin \theta_0 & \phi \end{bmatrix}$$

$$\frac{\partial^2 \Delta L_1}{\partial \theta^2} = 4L \cos \theta \rightarrow [K_{\Delta L_1}] = k_1 \Delta L_0 \begin{bmatrix} 4L \cos \theta & 0 \\ 0 & 0 \end{bmatrix}$$

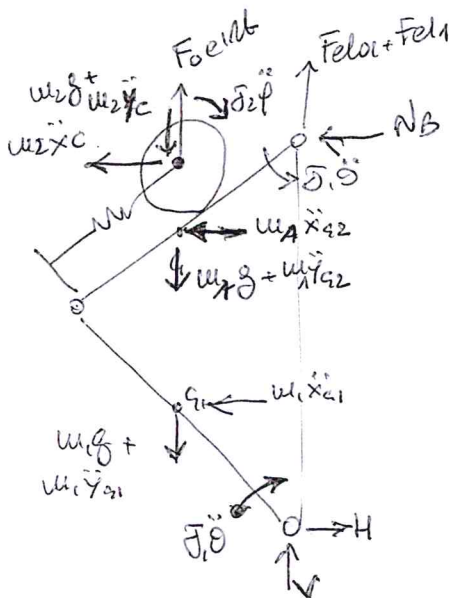
$$h_{q1} = y_{q1} \rightarrow \frac{\partial^2 h_{q1}}{\partial \theta^2} = -L \cos \theta \rightarrow [K_{q1a}] = m_1 g \begin{bmatrix} -L \cos \theta & 0 \\ 0 & 0 \end{bmatrix}$$

$$h_{q2} = y_{q2} \rightarrow \frac{\partial^2 h_{q2}}{\partial \theta^2} = -3L \cos \theta \rightarrow [K_{q1b}] = m_1 g \begin{bmatrix} 0 & 0 \\ -3L \cos \theta & 0 \\ 0 & 0 \end{bmatrix}$$

$$h_{q3} = y_c \rightarrow \begin{cases} \frac{\partial^2 h_{q3}}{\partial \theta^2} = -2L \cos \theta - (L_0 + x) \cos \theta - R \sin \theta \\ \frac{\partial^2 h_{q3}}{\partial \theta \partial x} = -\sin \theta \\ \frac{\partial^2 h_{q3}}{\partial x^2} = 0 \end{cases}$$

$$[K_{q20}] = m_2 g \begin{bmatrix} -(2L + L_0) \cos \theta - R \sin \theta & -\sin \theta \\ -\sin \theta & 0 \end{bmatrix}$$

• question 6 :



$$\sum F_y = 0: V - m_1 g - m_1 \ddot{y}_{q1} - m_1 g - m_1 \ddot{y}_{q2} - m_2 g +$$

$$- m_2 \ddot{y}_c + F_0 e^{i\omega t} + T_{el01} + T_{el1} = 0$$

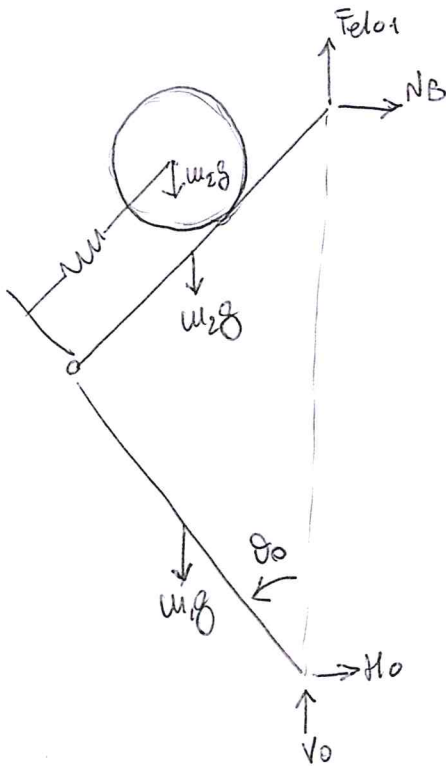
$$V = V_{ST} + V_{dyn}$$

$$V_{ST} = m_1 g + m_1 g + m_2 g + T_{el01}$$

$$V_{dyn} = m_1 \ddot{y}_{q1} + m_1 \ddot{y}_{q2} + m_2 \ddot{y}_c - F_0 e^{i\omega t} - T_{el1}$$

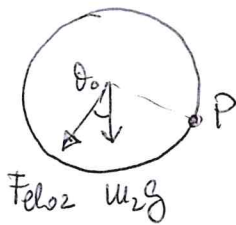
$$T_{el1} = (k_1 + i2c_1) \Delta L_1 \quad ; \quad \Delta L_1 = y_c e^{i\omega t} - y_3$$

• question 7 :



$$\begin{aligned} \sum M_B = 0 : & N_B 4L \cos \theta_0 - w_1 g 2L \sin \theta_0 + \\ & - w_2 g [R \cos \theta_0 + (2L - L_0) \sin \theta_0] = 0 \rightarrow \\ \rightarrow N_B = & \frac{2w_1 g L \sin \theta_0 + w_2 g [R \cos \theta_0 + (2L - L_0) \sin \theta_0]}{4L \cos \theta_0} \end{aligned}$$

$$\begin{aligned} \sum M_A^{AB} = 0 : & F_{el01} 2L \sin \theta_0 - N_B 2L \cos \theta_0 - w_1 g L \sin \theta_0 + \\ & - w_2 g (L_0 \sin \theta_0 - R \cos \theta_0) = 0 \rightarrow \\ \rightarrow F_{el01} = & \frac{2L N_B \cos \theta_0 + w_1 g L \sin \theta_0 + w_2 g (L_0 \sin \theta_0 - R \cos \theta_0)}{2L \sin \theta_0} \end{aligned}$$



$$\begin{aligned} \sum M_P^{disk} = 0 : & F_{el02} R + w_2 g R \cos \theta_0 = 0 \rightarrow \\ \rightarrow F_{el02} = & - w_2 g \cos \theta_0 \end{aligned}$$

$$\Delta l_{01} = \frac{F_{el01}}{k_1} \quad , \quad \Delta l_{02} = \frac{F_{el02}}{k_2}$$